

Why Meaning Emerges Locally in Finite Systems

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Uncle Phil and Aunt Viv. Julius and Rochelle. Red and kitty. Across generations of American television, family sitcoms have offered archetypes of healthy romance, stability, and partnership. These relationships all serve as role models, demonstrating balance through consistency, compromise, and communication. However, none of these compares energetically to the relationship between Hal and Lois in *Malcolm in the Middle*. In Season 4, Episode 6, "Forbidden Girlfriend," Hal and Lois are forced to abstain from sex for a week due to Lois being placed on medication.

Contrary to expectation, this period of sexual abstinence leads to a dramatic increase in household productivity. It turns out that when channeled, the energy once used for sex now gives way to order. Hal and Lois become productivity machines, completing tasks throughout the house almost mechanically. From a functional perspective, the benefits are evident, and it would seem that continued abstinence is the most optimal arrangement for continued productivity and efficiency. But when Lois' medication ends and intimacy resumes, the household collapses back into patterns of dysfunction. What used to be productivity, order, and efficiency returns to chaos, dysfunction, and instability. However, an important distinction must be made. Their regression is not indicative of failure. Instead, it feels like a restoration of wholeness. What returns with intimacy is not efficiency, but vitality. This phenomenon raises a question that applies beyond the domain of marriage. Why does constraint increase output at the sacrifice of aliveness? Why does release restore vitality while ignoring productivity? And why does the impulse to impose order feel in tune and correct even when it produces systems, moments, and experiences that feel bleak, lifeless, and unsustainable? This moment of coherence between Hal and Lois illustrates not just a lesson about romance but also a pattern observed in complex systems under pressure. Optimization through constraints can temporarily improve productivity but comes at the cost of adaptability, regulation, and vitality (Prigogine & Stengers, 1984). This event in the marriage of Hal and Lois is simply a depiction of a tension that exists throughout systems in our universe. There is a tension between energy control and release that must be compromised rather than commanded. This situation provides an entry point for understanding and observing the mechanics behind how energy moves, concentrates, and dissipates within finite systems.

Thermodynamically, the universe is framed as a closed system, an isolated system with no external energy source to act upon (Schrödinger, 1944). The caveat is that the universe is not static or uniform; energy is neither added to nor removed from it. Energy is only redistributed, as there's no external point to stabilize what breaks down nor restore what is lost. Entropy describes the natural inclination for energy to disperse and distribute among all available states within an evolving system (Frigg & Werndl, 2016). Within this, entropy is not the force pushing the universe towards decay or an accumulated substance as it's commonly understood, but instead a statistical description of the capacity of energy for change. This dispersion leads to inherent irreversibility: although systems do not collapse instantly, certain initial configurations become

statistically improbable over time because the conditions required to reconstruct them no longer exist (Earman, 2006).

The consequence of this is that all structure within this system exists under constraint. Every system in the universe is an open system, with living organisms such as human beings existing as open systems within a closed totality (Schrödinger, 1944). An open system remains one through continuous exchange and flow (Morowitz, 1968). Energy, matter, and information must be able to flow in and out of their boundaries for continuous existence. Without the intake of resources, output of waste, and regulation, the system moves towards isolation and the collapse of its unique structure. Therefore, life and entropy coexist through the local management of entropy flow. The exportation of entropy is crucial to the survival of any open system, as metabolism, respiration, waste, communication, and behavior all function as mechanisms for redistributing energetic load outward so that internal structure can temporarily persist (Schrödinger, 1944). As stated earlier, every system in the universe exists under constraint, so the apparent stability of a living system is not static but an ongoing process in which its structure is continuously rebuilt, repaired, and reconfigured in response to external change and internal insufficiency (Kauffman, 1993).

Humanity has attempted to frame this tension between release and order through many lenses. Morally, relationally, and psychologically. These frameworks continue to fall short by burying the phenomenon in values and labels before acquiring true insight into it. However, I urge you to analyze this from a thermodynamic perspective. In thermodynamics, entropy describes the statistical tendency of energy to spread across available states. It can be observed that Lois and Hal's abstinence imposed more constraint on the system, resulting in higher output. This is observed because constraints reduce the degrees of freedom. When variability is suppressed, energy is forced into narrower channels, producing short-term efficiency at the expense of adaptability, as the energy lacks sufficient capacity to move among different states. Systems appear orderly because the range of possibilities has been minimized. The amount of uncertainty has been reduced at the expense of the system's capacity to respond to change. In contrast, release restores degrees of freedom by allowing variability to be explored as energy is spread throughout the system. This, of course, undermines optimization, as with the return of vitality, the order once directed for a single outcome is sacrificed.

In some societal frameworks, sex is often associated with the scattering of energy, the reduction of order, and the decomposition of structure. Yet this framing allows for a more honest investigation; one in which neither sex nor entropy is inherently productive or destructive. Both function as mechanisms of energy redistribution. Whether they result in productivity or destruction depends not on the process itself but on the structure through which energy flows. Thermodynamically, structure refers to the organization of constraints that channel energy into work rather than dissipate it. Entropy and work are not inversely related; work is a temporary state enabled by gradients, timing, and capacity (Nicolis & Prigogine, 1977). When energy is introduced into a sufficiently structured system, the structure enables the energy to be converted into organized action. The same energy could be introduced in a system at a different time, with

a smaller capacity, or without sufficient structure; it manifests as dissipation. An important distinction is that, in both cases, neither outcome is intrinsic to the energy itself.

Despite this, humanity continues to misinterpret this relationship, consistently struggling to maintain the balance between constraint and vitality. Though one can point fingers at specific frameworks and hold them responsible, this misinterpretation does not stem solely from ignorance or lack of intelligence. Rather, we must examine the structure of the system through which we are even able to perceive the gradient. Our cognition evolved for moments where rapid decision-making was needed. Where one inkling was the difference between survival and death. Our models of reality were based simply on what line of decision would enable me to survive and my genes to proceed. Models like Hoffman's Fitness Beats truth illustrate this theory, as it states (Hoffman, 2014). The current perspective humans have of reality is more conducive to a model that allows me to live the longest and procreate than to objective observation. Accuracy is expensive, and evolution is astigmatic. As part of this, cognitive features that enabled survival and reproduction were subsequently passed on, even at the cost of simplifying, distorting, or compressing reality. As such, this is most evident in the cognitive biases that confer speed, safety, and social advantage. Mental shortcuts, or heuristics, evolved toward loss aversion, heightened threat sensitivity, pattern completion, and categorical thinking to reduce computational load associated with ambiguity and uncertainty (Tversky & Kahneman, 1974). The result of this compression is a systemic mismatch between how reality unfolds objectively and how it is experienced subjectively. Dynamic processes are rendered as linear sequences with clear causes and effects, and start and end points. Continuous tradeoffs are perceived as binaries. Now, this is to be taken with a grain of salt, as over millions of years of human evolution, these systems clearly did their job. But it could be that perception should not be taken at face value, but instead be understood as an interface for control tuned by the flow of information required for action (Hoffman, 2014; Fields, 2019).

Humanity stands at a crossroads, pulled between perceptions of extremes with no consistent equilibrium. Our perception offers a medium for experiencing reality, but it lacks the depth to fully understand it. As the active inference framework details, what we call perception is simply an approximation (Friston, 2010). Inference itself is reformulated as prediction and even then as approximation. The agent must operate on simplified internal models, updating "posterior probability distributions over all non-observed random variables" rather than directly accessing the world (Friston, 2010). Exact observation of the universe is not feasible beyond the simplest of systems, which means approximation is a necessity, not a failure. Regulation cannot be imposed from some supreme view, but instead it is something that is negotiated through motion. Therefore, the issue does not lie in choosing one extreme over the other, but in learning how the system itself updates as conditions shift. The conflict between constraint and vitality cannot be solved by any one rule, ideology, or permanent arrangement because the conditions that allow for stability are themselves in motion. Since energy is always moving, and structure is temporary, the only place this conflict can be understood is not through abstraction, but in experience. What appears optimal in one moment just as easily becomes unsustainable in the next. We do not have

the ability to instantly rewire our evolutionary predispositions. We can observe the loop that allows us to observe, process, and form interpretations about the world around us. In the process through which perception becomes action, and action reshapes perception, we must examine the loop itself (Clark, 2016).

The loop begins in sensation, which feeds data into a model. The nervous system doesn't perceive the world in its entirety; instead, it receives electrical signals constrained by receptors, filtered by attention, and shaped by prior expectation. What arrives as experience has already been narrowed by biology and predictive bias. The interpreter then converts that signal into relevance. From a spectrum of varying degrees, the mind extrapolated threat or safety, gain or loss, and attraction or aversion. Impulse emerges as the body's preparation to act on that interpretation. It is the mediator between cognition and expenditure, motivation that reflects the system's attempt to reduce uncertainty or secure stability (Friston et al., 2017; Hohwy, 2013).

Action is the moment where abstraction becomes tangible. Energy is spent. Alternatives close. The environment is altered. Feedback follows as the world returns information on the consequences of that choice. The intermanal model is adjusted. Predictions are refined or destabilized, and structure shifts incrementally. This loop is regulated in its only emergent form. There is no external vantage point to which perfect order can be imposed. There is only this exchange between the organism and the environment.

This extrapolation of Hal and Lois was never a lesson in romance but in regulation. The brief surge of productivity under constant and return to disorder with intimacy were not opposites of success and failure, but instead phases in the oscillation of an open system managing energy in a closed universe. Constraints narrow degrees of freedom and procure rigidity, while release restores variability and revives vitality. Neither state is permanent, and neither is inherently correct. Entropy doesn't dictate decay, and work doesn't inherently dissipate structure. Both, instead, are processes of energy redistribution whose consequences depend on timing, capacity, and configuration. Our ability to perceive these consequences evolved for survival rather than precision, compressing these gradients into binaries and seeking fixed equilibria feedback within the perception-action loop. Sensation narrows, interpretation extracts relevance, impulse prepares expenditure, action closes alternatives, and feedback reshapes structure. Meaning arises precisely there, where finite systems act under inverse constraints and feel consequences (Barrett, 2017). In a universe where structure is temporary and equilibrium is dynamic, the task is not to choose once and for all between order and release but to regulate their rhythm in motion.

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